

Pawsify

Parenting and Second Chances

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Abstract

“Every pet deserves **love**, and every street dog deserves a **chance**.” That’s the vision behind this project.

Pawsify isn’t just an **app**; it’s a step toward making pet care simpler, more responsible, and more compassionate. This platform brings together pet owners, **veterinarians**, and animal welfare organizations to make those challenges easier.

Beyond pet ownership, this project focuses on rescue, foster care, and street dog welfare by building a strong **network** of animal lovers, vets, and rescue organizations helping lost, abandoned, and injured animals. Through spreading awareness, community support, and responsible ownership, this project aims to reduce pet abandonment and improve overall animal **welfare**.

When people come together to take responsibility, show kindness, and take action, **we don’t just help animals, we change their lives.**

Motivation

We envision making **pet care** easier and **rescue** efforts stronger. Many pet owners struggle to keep track of vaccinations, find trusted services, and connect with a supportive **community**. At the same time, street dogs continue to suffer due to a lack of rescue and **foster care**. Even those who want to help often don't have access to the right **resources**.

My team and I are creating **Pawsify** a platform designed to simplify pet care, support rescue initiatives, and bring together pet parents, veterinarians, and animal welfare organizations.

A world where every pet is cared for, every rescue gets a second chance, and no animal is left behind.

Key Features

- **User & Dog Profiles** — Create and manage personal and dog profiles with detailed information and photos.
- **Playdate Functionality** — Browse, filter, and schedule nearby dog playdates with map-based location support.
- **Matchmaking Functionality** — Get smart matches for companionship or breeding based on compatibility and preferences.
- **ResQ Hub** — Access adoption, foster care, donations, rescue collaborations, and missing/found reporting.
- **Health & Care Management** — Track vaccinations, health records, and breed-specific care recommendations.
- **Locator Search** — Find nearby shops, parks, grooming centers, and veterinary clinics using map integration.
- **Lost & Found Reporting** — Submit and browse reports of lost or found dogs with photos and locations.
- **Events & Volunteering** — Discover local dog-related events and volunteer opportunities.
- **Checklist on Home Page** — Quick access to essential tasks and reminders for dog care.
- **Notifications** — Receive alerts for matches, events, and updates.
- **Safety & Verification** — Ensure secure interactions with verified profiles and reporting features.

Hardware & software Specifications

Hardware Specifications		
Development Environment	Deployment Environment	Client Environment
Intel Core i5-12450H / Apple M2 16 GB DDR4 RAM 1Tb SSD Windows 11 / MacOS	Intel Xeon 16GB RAM 100GB+ SSD Windows OS	Snapdragon 665 / A11 Bionic or better Minimum 3GB Android 8.0+ / iOS 14+ At least 350MB free space

Software Specifications		
Development Environment	Deployment Environment	Client Environment
Flutter SDK v3.27.2 Firebase v13.30.0 Firebase v9.4.2 Firebase Auth v4.19.0 Google Maps API v3.59.8	AWS Lambda Cloud Firestore DDoS Protection RESTful APIs	Flutter-based app API Communication Local Storage for Cache Encrypted Data Storage Location & Notifications

Functional Requirements

User Authentication: Secure login & role-based access via Firebase

Profile Management: Manage user and pet profiles with health records

Health Tracker: Log vaccinations/deworming with reminder notifications

Educational Content: Breed-specific tips via APIs & in-app content

Dog Matchmaking: Connect pets for playdates & mating based on preferences

Rescue & Foster: View/adopt dogs listed by NGOs or individuals

Payments: Secure donations & fees via Razorpay

Location Services: Google Maps nearby rescues

Non-Functional Requirements

Performance: Real-time, smooth user experience using Firebase

Scalability: Built to handle growing users with Firestore & AWS

Security: Encrypted data, secure access, DDoS protection

Usability: Flutter ensures clean, cross-platform UI

Compatibility: Android 8.0+ / iOS 14+, 3GB RAM, 350MB storage

Responsive Design: Optimized for all screen sizes and orientations

Real-world Applications

- **Hostel & Day Care** – This feature helps pet parents find trusted boarding homes or daycares for their dogs when in travel or busy, thus they could be in safe hands while their owners are away.
- **Rescue, Adoption & Foster Care** – Connects rescue organizations with prospective foster homes which will provide a safe place for rescued or abandoned dogs until they find permanent homes, freeing the already overfull shelters of their burden, donate or volunteer to support indie dogs in your area.
- **Vet/Grooming & Shop Locator** – This feature makes it easy for dog owners to locate veterinary clinics, emergency vet care, grooming services and pet shops for pet necessities.
- **GPS Tracking** – Enables real-time location tracking of dogs thereby ensuring that an owner does not lose the track of his pet, while enabling an easy and quick tracking in case a pet is lost.
- **Emergency SOS** – The SOS button used by the owner can be pressed to inform the nearby vet, pet shelter, or fellow users of the emergency situation of accident or medical emergencies for urgency of help being rendered. It also helps in reporting dog abuses.

Background **study in the domain**

Dogs are everywhere as pets in homes or strays on the streets, but many live in difficult conditions. Abandonment, abuse, and health issues affect countless dogs, with little access to proper care. Even well-meaning pet parents often struggle to find trusted veterinary, grooming, or emergency services.

In India, there is a significant lack of awareness about proper dog nutrition, socialization and behavior training, as well as responsible adoption and mating practices, which often leads to preventable health, behavioral, and welfare issues among both pets and strays.

Stray dogs, meanwhile, suffer from limited access to food, shelter, and medical care, with rescue efforts often scattered and hindered by poor coordination and low public engagement.

OUR VISION:

There is a clear need for a unified and accessible platform that brings **together help, care, support, education, and community**, creating one connected space dedicated to improving the lives of dogs.

Tools and Technologies Survey

Flutter (with Dart) – For building UI as it supports multi platform using single code.

Firebase Firestore – For managing the database and backend.

Firebase Auth – For login and sign up for the app.

Firebase Cloud Messaging – For reminders and alerts.

Firebase Analytics – For tracking user behaviour and app performance.

Razorpay – For secure payments.

AWS – For managing users smooth experience on the app.

Restful APIs – For API integrations

I5 12450H / M2 – Smoother and faster processing.

16GB RAM – For better multi tasking.

1TB SSD – For faster data transfers

Research Paper 1

“Critical Study on Animal Cruelty in India”

Published in: 2021, International Journal for Research in Engineering Application & Management (IJREAM)

Key Findings:

The paper highlights the ongoing issue of animal cruelty in India, especially towards dogs. It points out weak law enforcement, low public awareness, and lack of strict punishment as major challenges. NGOs try to help but face many limitations.

How It Helps Pawsify:

This paper supports our goal to create awareness, build community support, and offer tools like reporting and rescue options through our app to help reduce cruelty and improve dog welfare.

Research Paper 2

‘Stray appetites’: a socio-ecological analysis of free-ranging dogs living alongside human communities in Bangalore, India

Published in: December 2021, Urban Ecosystems

Key Findings:

Feeding by households and bakeries is the main driver of stray dog populations, more than garbage or housing density. Just 10–20% of feeders can sustain large population. Public awareness of sterilization is low, and feeding often leads to territorial behavior and conflict..

How it helps Pawsify:

The study supports Pawsify’s features like ABC tracking, feeding logs, and hotspot mapping. It reinforces our mission to combine tech with education and community efforts to promote responsible stray care and reduce human-dog conflict.

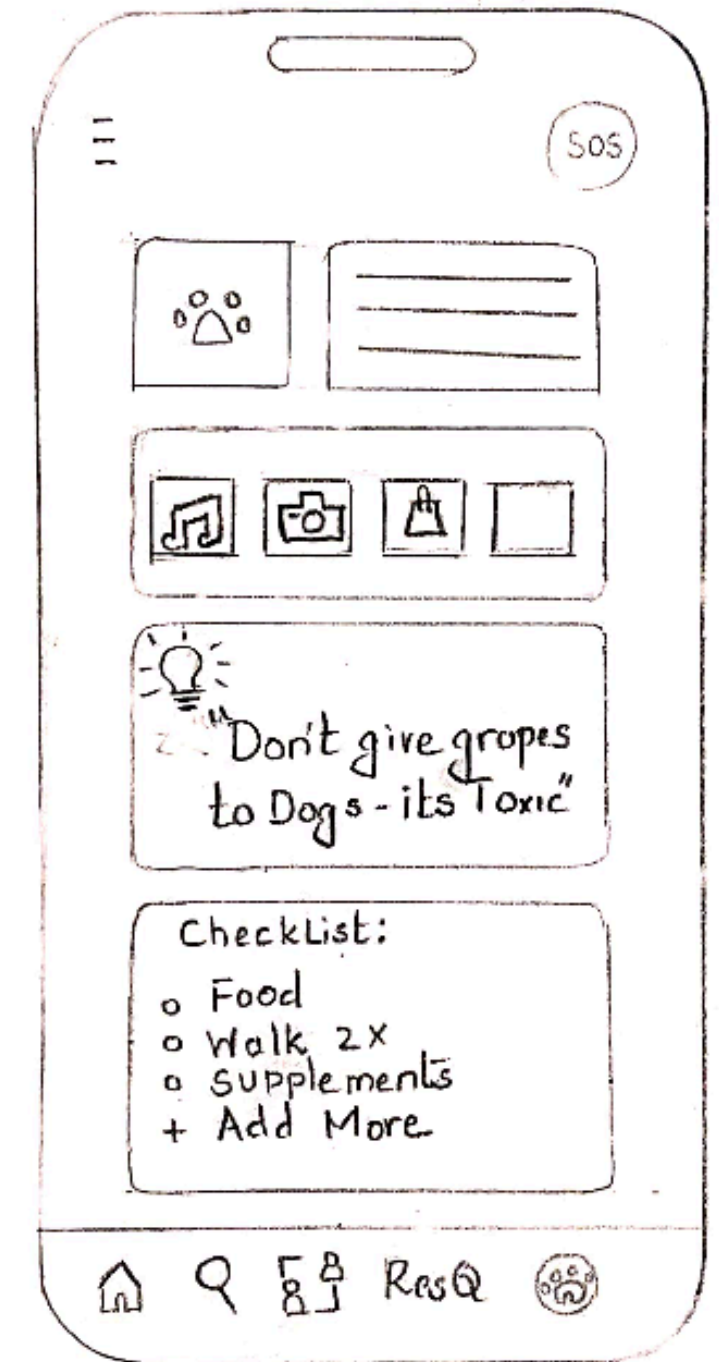
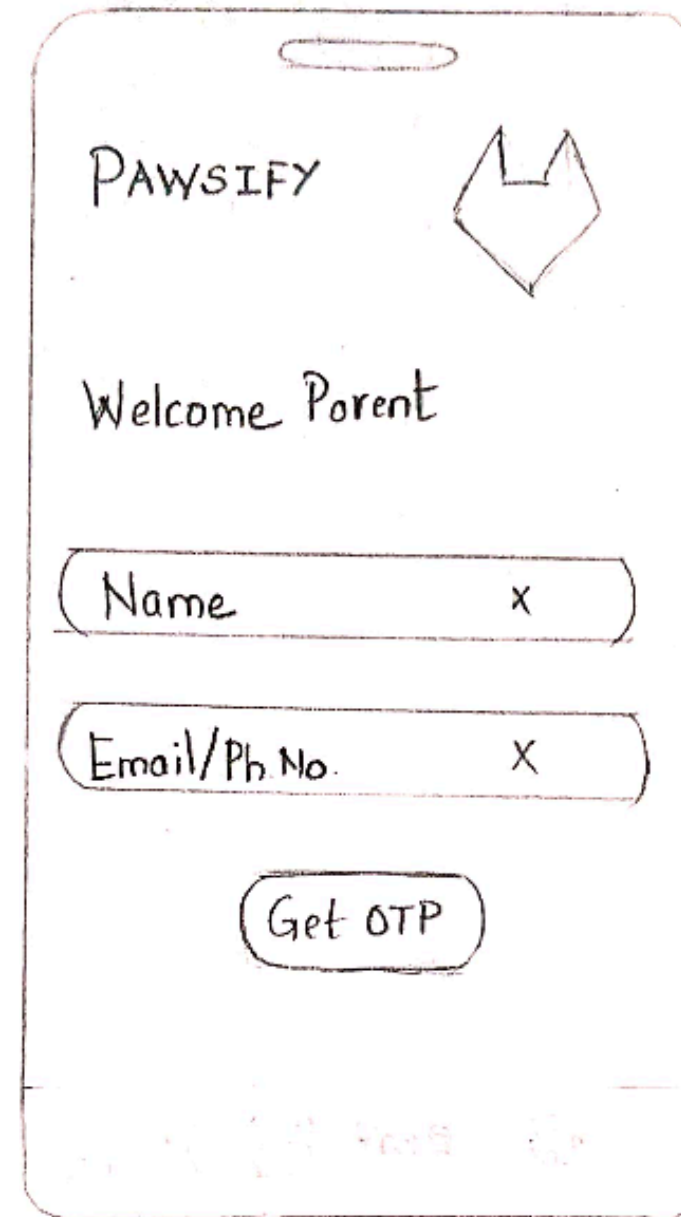
Object and Scope

The project aims to create a **user-friendly mobile platform** that streamlines pet parenting. It helps dog owners manage **vaccinations, health records, and provides GPS tracking and emergency alerts** to enhance pet safety. The app also promotes ethical breeding and encourages responsible pet ownership.

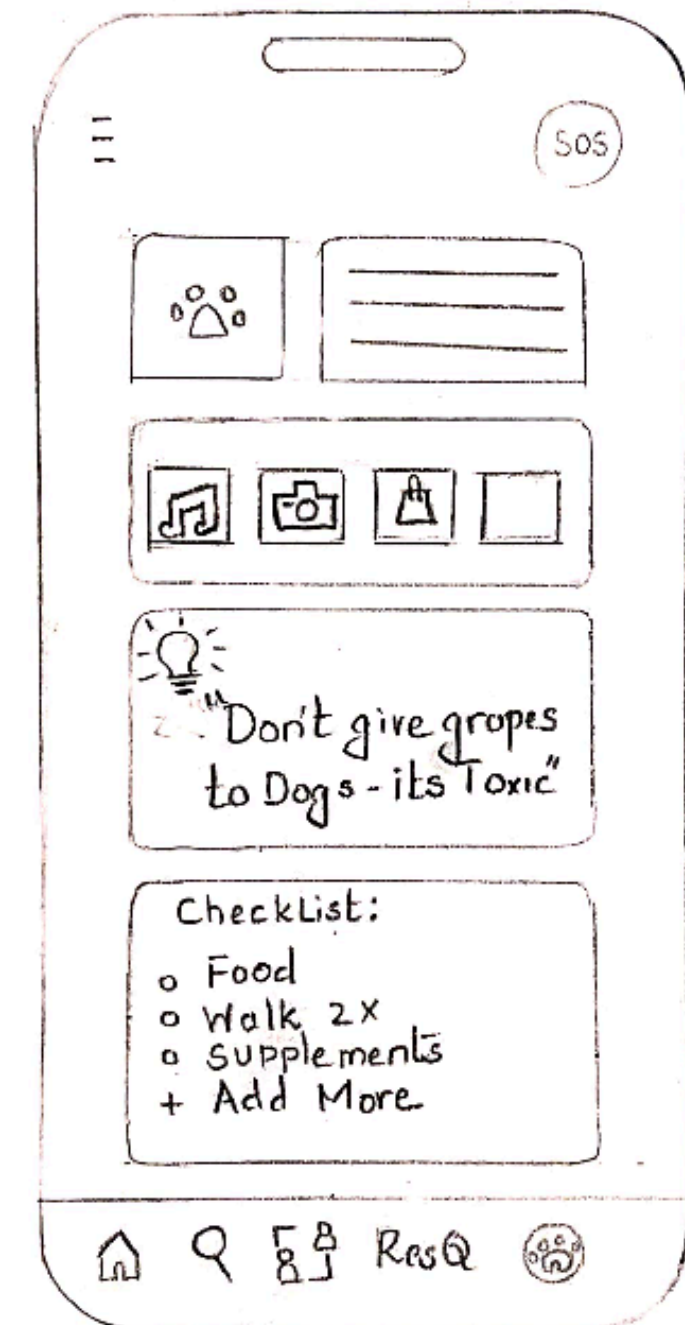
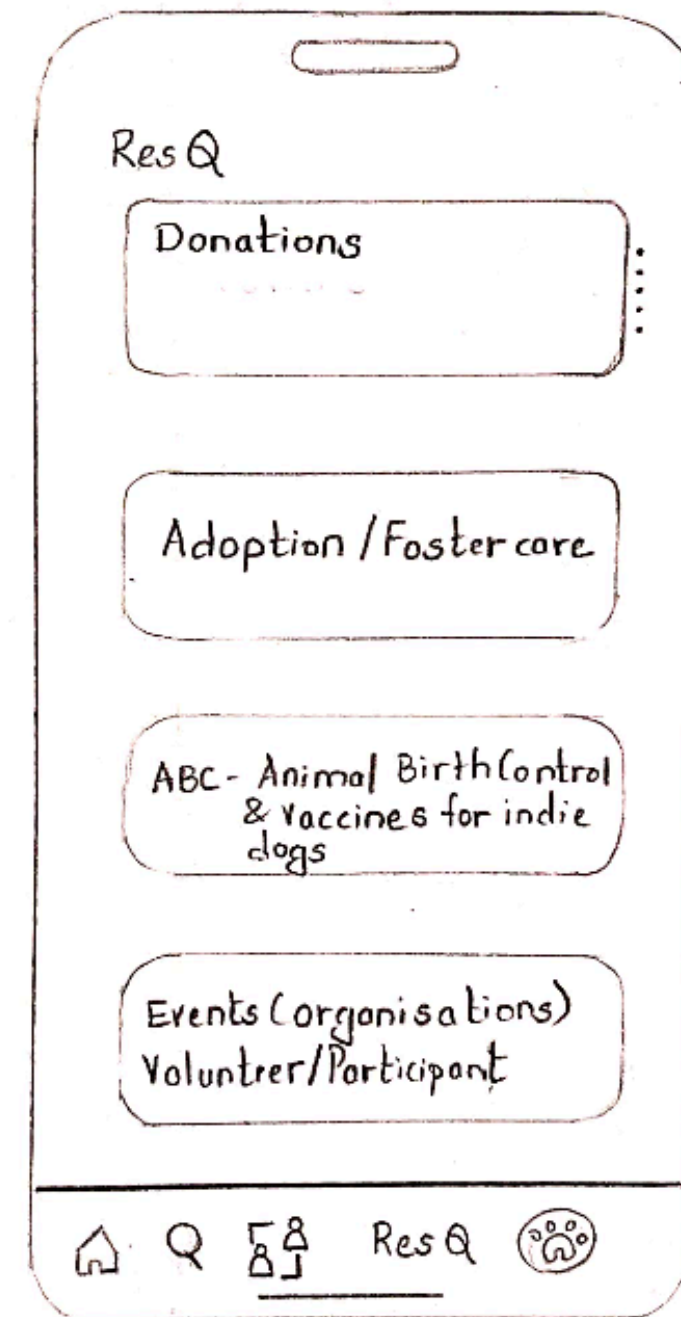
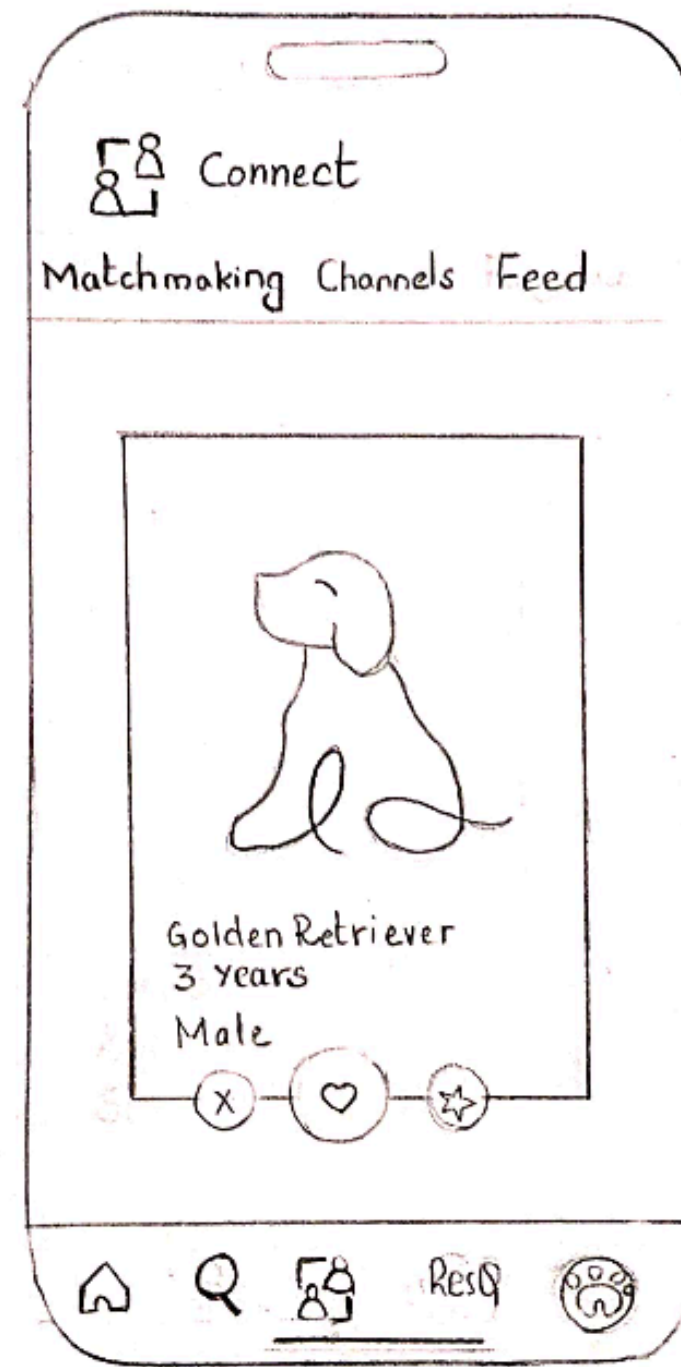
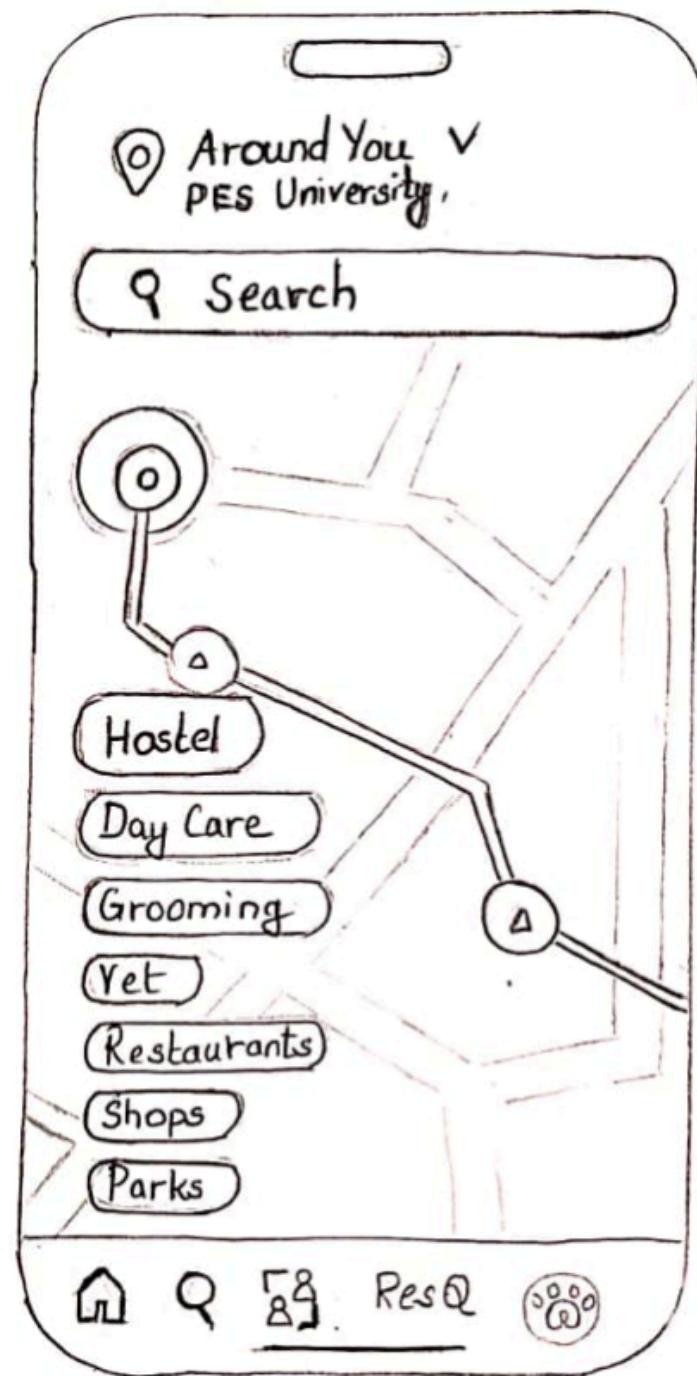
Beyond routine care, the platform builds a complete ecosystem for dog owners. It connects users with **animal shelters, NGOs, and foster** networks to support rescue and adoption. It also simplifies access to trusted veterinarians and groomers.

Available on **Android** and **iOS**, the app supports real-time syncing and offline access. By combining **health tracking, safety tools, and community engagement**, it delivers a holistic solution for modern pet care.

Sketch Preview



Sketch Preview



Project Description

In India, many dogs, both **pets** and **strays**, face issues like **abandonment**, **abuse**, and lack of proper care. Pet parents often struggle to find **reliable veterinary**, **grooming**, and **emergency services**, while stray dogs suffer without **food**, **shelter**, or **medical help**. Unregulated **crossbreeding practices** also lead to health complications in many breeds.

This platform aims to support **animal welfare organizations** by offering a **centralized space** for **services**, **education**, and **community support**. It promotes **responsible adoption**, proper care, and **awareness** on topics like **dog nutrition**, **socialization**, and the dangers of crossbreeding, ultimately improving the **well-being of dogs** across the country.

Review of Existing Systems

1. Kuddle

Focuses on pet services such as grooming and vet care with a social network for pet parents. Lacks rescue support, mood or symptom tracking, and real community impact.

2. Heads Up For Tails

Offers a strong online store for pet products and wellness items. Mainly a shopping platform without community features or emergency help.

Why Pawsify is Better

Combines care, community, and technology by connecting pet parents with vets, shelters, and groomers while supporting rescue, foster, and adoption. Includes SOS alerts, GPS tracking, and health monitoring in a single unified platform.

Problem Statement

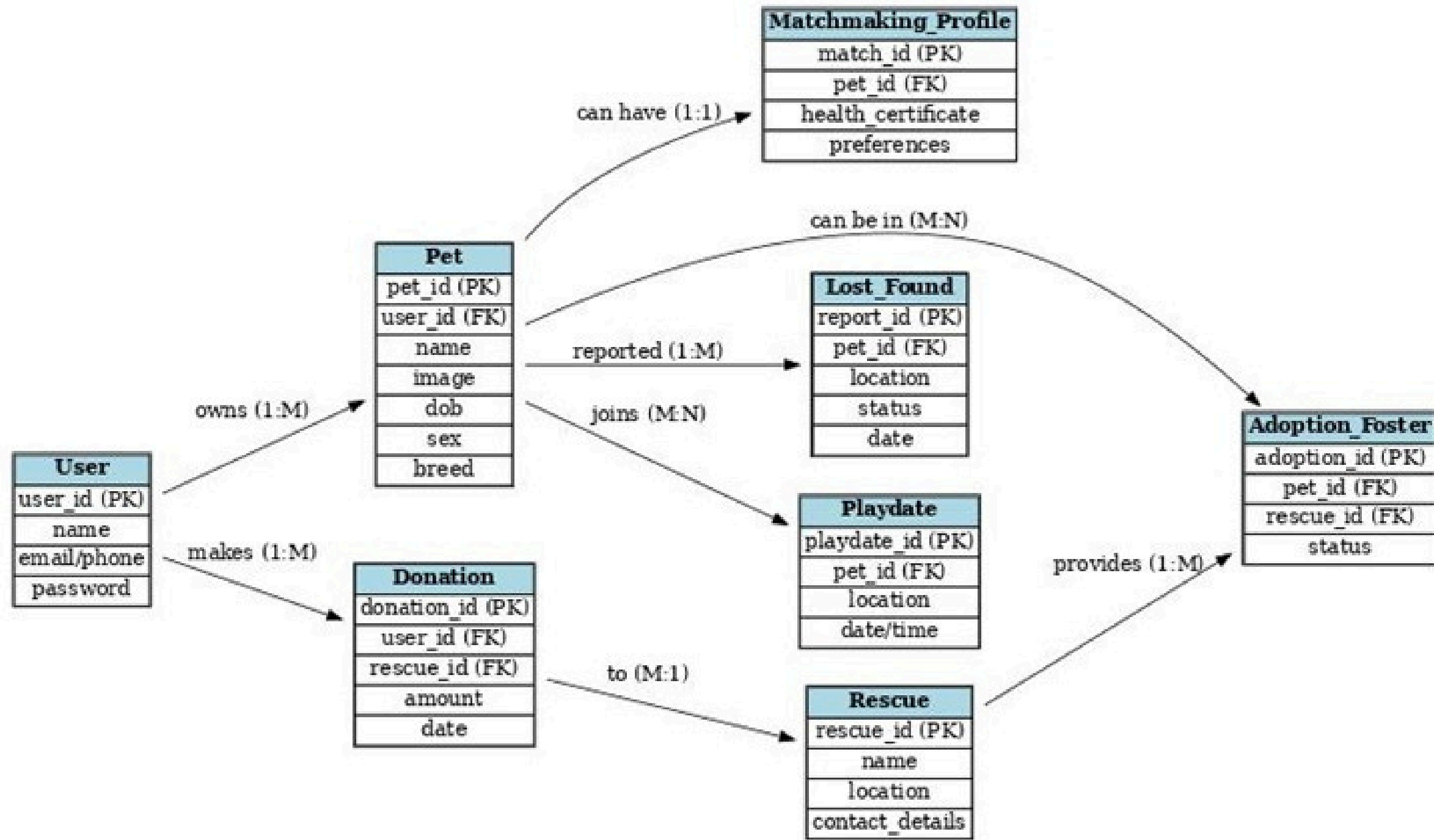
Despite the rise in **pet ownership**, there is still a big gap in proper care solutions that support both **family pets** and **street animals**. The problems faced are:

- Pet parents often miss **vaccinations, deworming, and diet schedules**.
This leads to **health risks and stress** for both pets and owners. Pawsify offers smart **reminders** and **health trackers**, making it easy to stay consistent and organized. It helps ensure pets get timely care without confusion.
- No single platform connects **owners, vets, and NGOs**.
This causes gaps in care and limited collaboration. Pawsify creates a unified space where pet owners, vets, and NGOs work together for better pet welfare. It builds a stronger support system for every pet.

Problem Statement

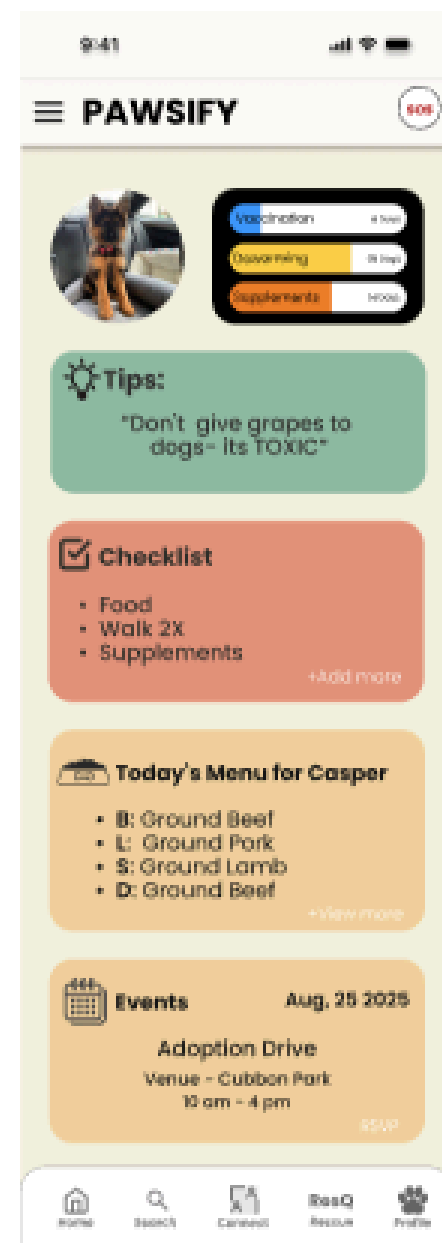
- Street dogs **lack rescue, foster care, and medical support.**
This leaves many strays without proper attention or safety. With NGO tie-ups, foster care features, and rescue reporting, Pawsify ensures better support for strays. It helps create a stronger network for animal welfare.
- Existing apps focus mostly on **shopping** and **services.**
They rarely address the real challenges of pet care and street animals. Pawsify focuses on **welfare** and **community, promoting awareness** and responsible pet parenting. It encourages meaningful impact beyond just services.
- Pet parents struggle to find trusted playmates or mates for their dogs.
This often limits socialization and healthy connections. Pawsify provides a safe dog matchmaker feature for **playdates** and **breed-specific** connections. It ensures pets enjoy companionship in a secure way.

Diagram



Demo of Project

home page



9:41

PAWSIFY

 **Vaccination**
Deworming
Supplements

Tips:
"Don't give grapes to dogs- its TOXIC"

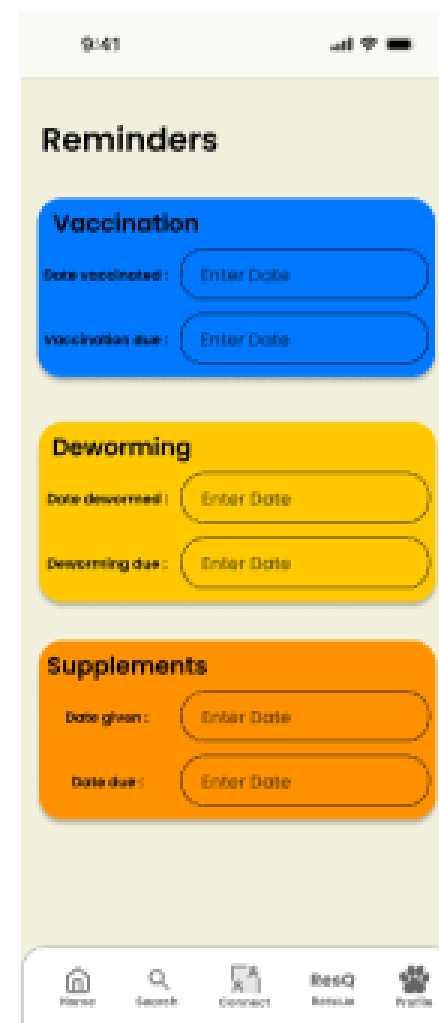
Checklist
• Food
• Walk 2X
• Supplements
[+Add more](#)

Today's Menu for Casper
• B: Ground Beef
• L: Ground Pork
• S: Ground Lamb
• D: Ground Beef
[+ View more](#)

Events **Aug. 29 2025**
Adoption Drive
Venue - Cubbon Park
10 am - 4 pm
[Join](#)

Home Search Connect ResQ Rescue Profile

edit



9:41

Reminders

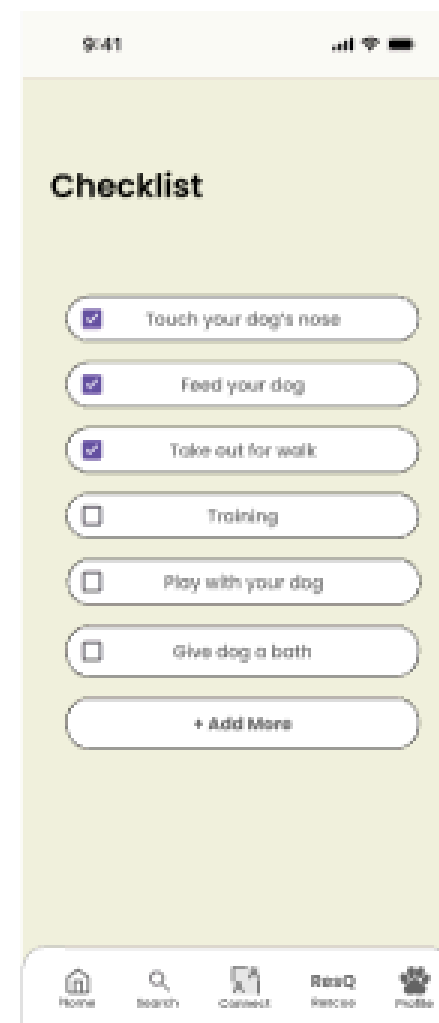
Vaccination
Date vaccinated:
Vaccination due:

Deworming
Date dewormed:
Deworming due:

Supplements
Date given:
Date due:

Home Search Connect ResQ Rescue Profile

h2



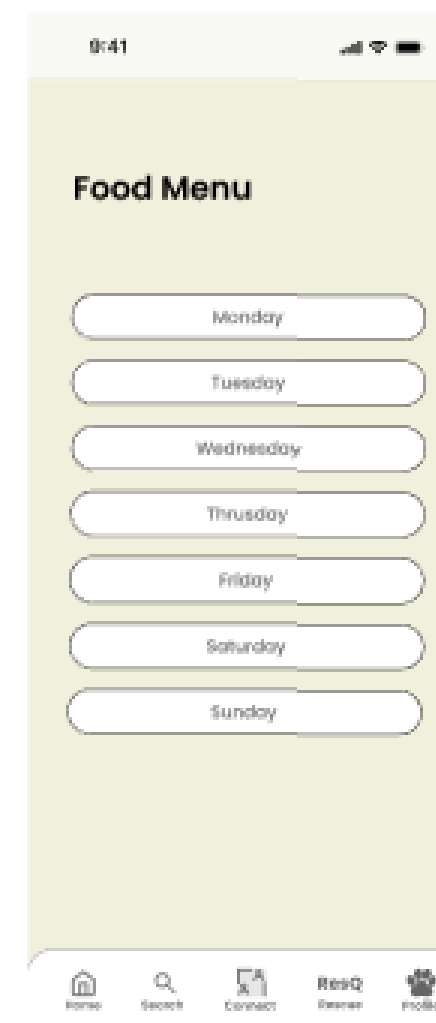
9:41

Checklist

☒ Touch your dog's nose
☒ Feed your dog
☒ Take out for walk
☐ Training
☐ Play with your dog
☐ Give dog a bath
[+ Add More](#)

Home Search Connect ResQ Rescue Profile

h3

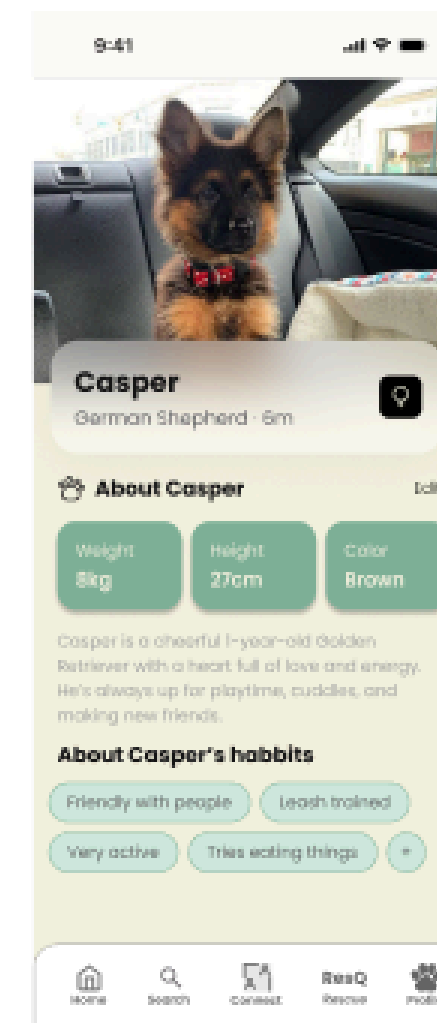


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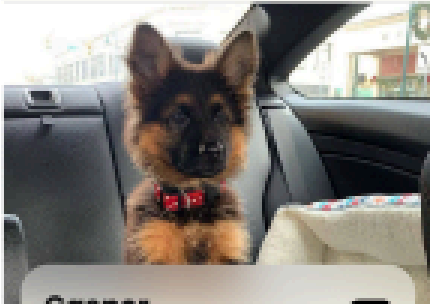
Food Menu

Home Search Connect ResQ Rescue Profile

pet profile



9:41



Casper
German Shepherd - 6m

About Casper [Edit](#)

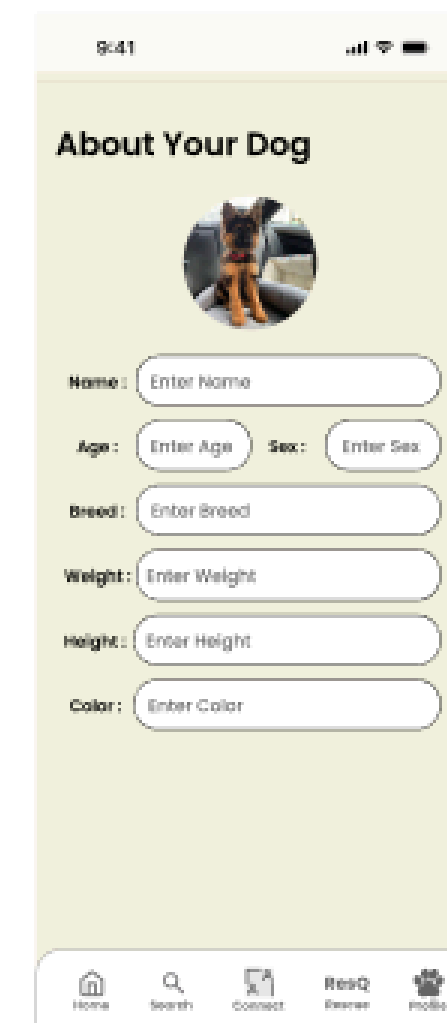
Weight: Height: Color:

Casper is a cheerful 1-year-old Golden Retriever with a heart full of love and energy. He's always up for playtime, cuddles, and making new friends.

About Casper's habits
☒ Friendly with people ☒ Leash trained
☒ Very active ☒ Likes eating things [+](#)

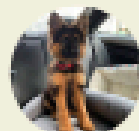
Home Search Connect ResQ Rescue Profile

edit



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About Your Dog

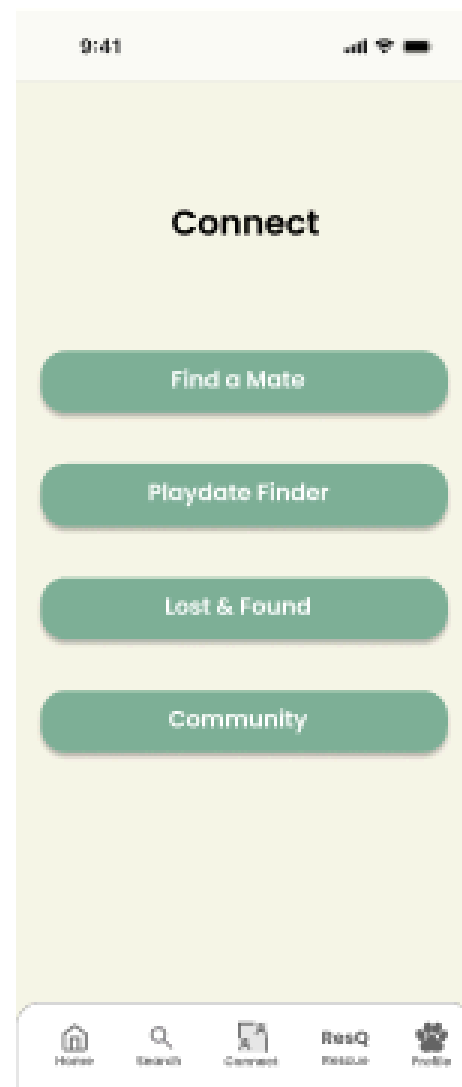


Name:
Age: Sex:
Breed:
Weight:
Height:
Color:

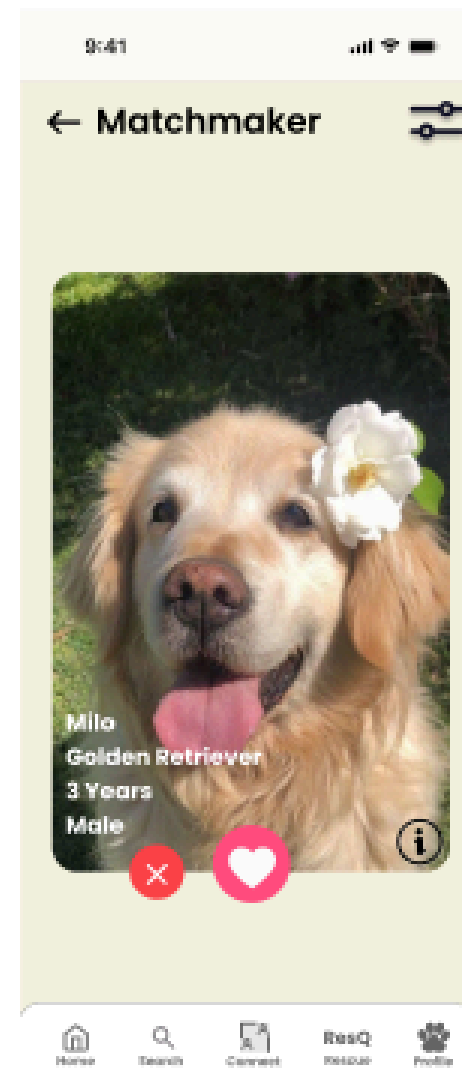
Home Search Connect ResQ Rescue Profile

Demo of Project

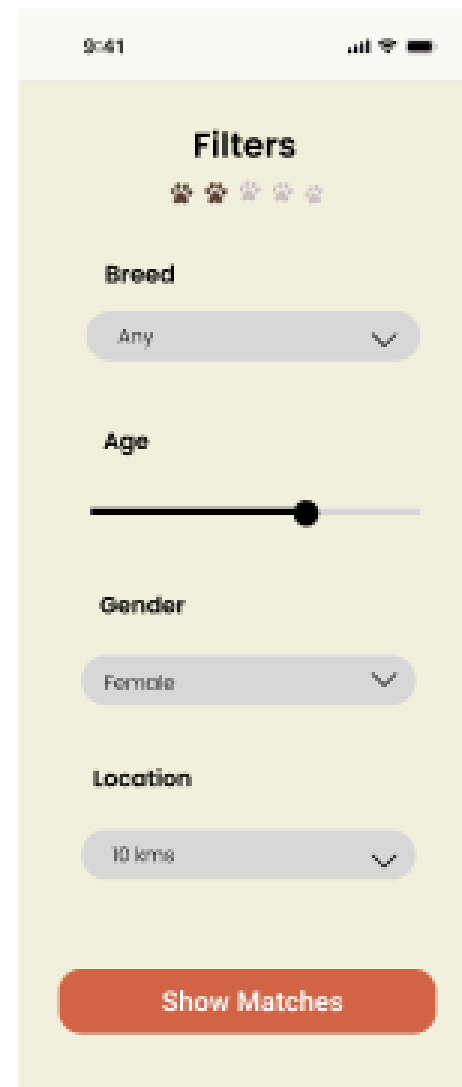
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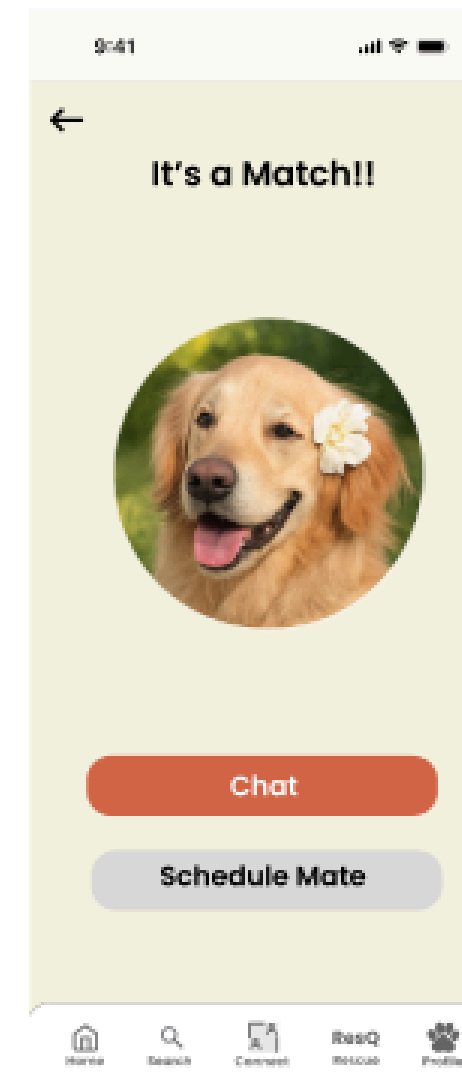
matchmaking



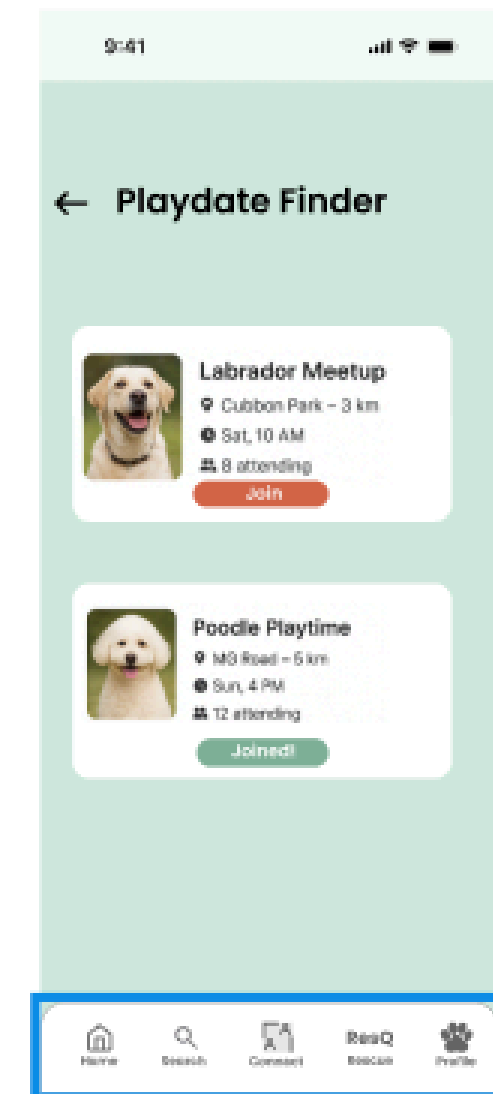
filter



match



p1



Challenges & Solutions

OTP Verification

Secure login setup for pet parents and NGO partners was a concern.

Implemented Firebase OTP authentication to ensure reliable and safe user verification.

Team Coordination

Testing across different devices created compatibility issues.

Solved this by testing on multiple Android versions and devices, and making necessary fixes for stability.

Connecting to Firebase

Connecting the app with Firebase database created difficulties.

Configured the SDK and used Firestore to handle real-time data storage and retrieval.

Testing on Different Devices

Team faced coordination issues due to different design ideas.

Followed an Agile approach with regular meetings and used Figma for smooth collaboration.

Project overview and Module Descriptions

Project Overview:

Comprehensive app for pet owners which combines:

- Pet profiles & health care
- Playdates & matchmaking
- Service locator: Shops, Parks, Vets, Groomers
- ResQ hub: Rescue, Adoption, Donations, Lost & Found
- Community Events & Volunteering.

Built with Firebase for secure authentication, real-time updates, and data storage.

Module Descriptions:

- **Authentication:** Secure login/registration via email or OTP; stores user & pet data.
- **Home:** Dashboard with reminders, checklists, menu, and pet care tips.
- **Search:** Map-based search for nearby pet services.
- **Connect:** Matchmaking, playdate finder, community hub, and events.
- **ResQ:** Rescue, adoption, donations, ABC events, lost & found reporting.
- **Profile:** Manage editable profiles with secure uploads.

Working System Modules

1. Authentication Module

Secure login/registration via email or OTP.
Stores user & pet details in Cloud Firestore.
Maintains session states.

2. Home Module:

Dashboard with reminders, checklists, food menus, and pet care tips.

3. Connect Module :

Matchmaking: Discover compatible pets.
Playdate Finder: Browse and schedule playdates.

4. ResQ Module (Demo Ready):

Lost & Found Reporting: Reports with photos and location details.

5. Profile Module (Demo Ready):

Manage user and pet profiles with editable details and uploads.

6. Search Module (In Progress):

Locator for nearby shops, parks, groomers, and vets (map-based search).

Testing Approaches

- Manual testing was carried out on the **Chrome emulator** to verify all core functionalities.
- A total of **11 test cases** were created and executed across all active modules.
- **Functional Testing:** Validated login, matchmaking, playdate, and profile management workflows.
- **Integration Testing:** Verified Firebase Authentication, Firestore, and Storage connections.
- **UI/UX Testing:** Ensured smooth navigation, consistency, and responsiveness.
- **Error & Crash Testing:** Checked for stability and proper handling of invalid inputs.
- **Bug Fix:** Resolved matchmaking session persistence issue by resetting session variables in Firebase sync.

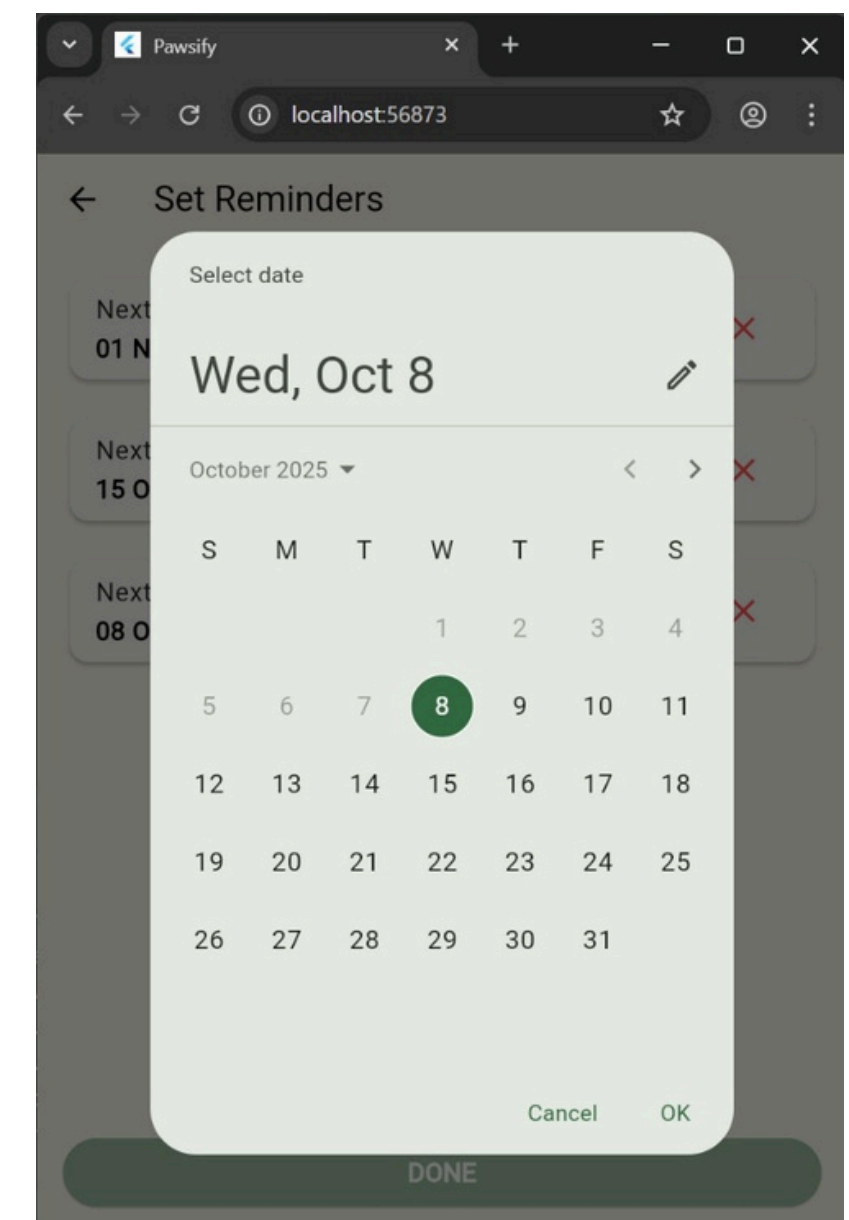
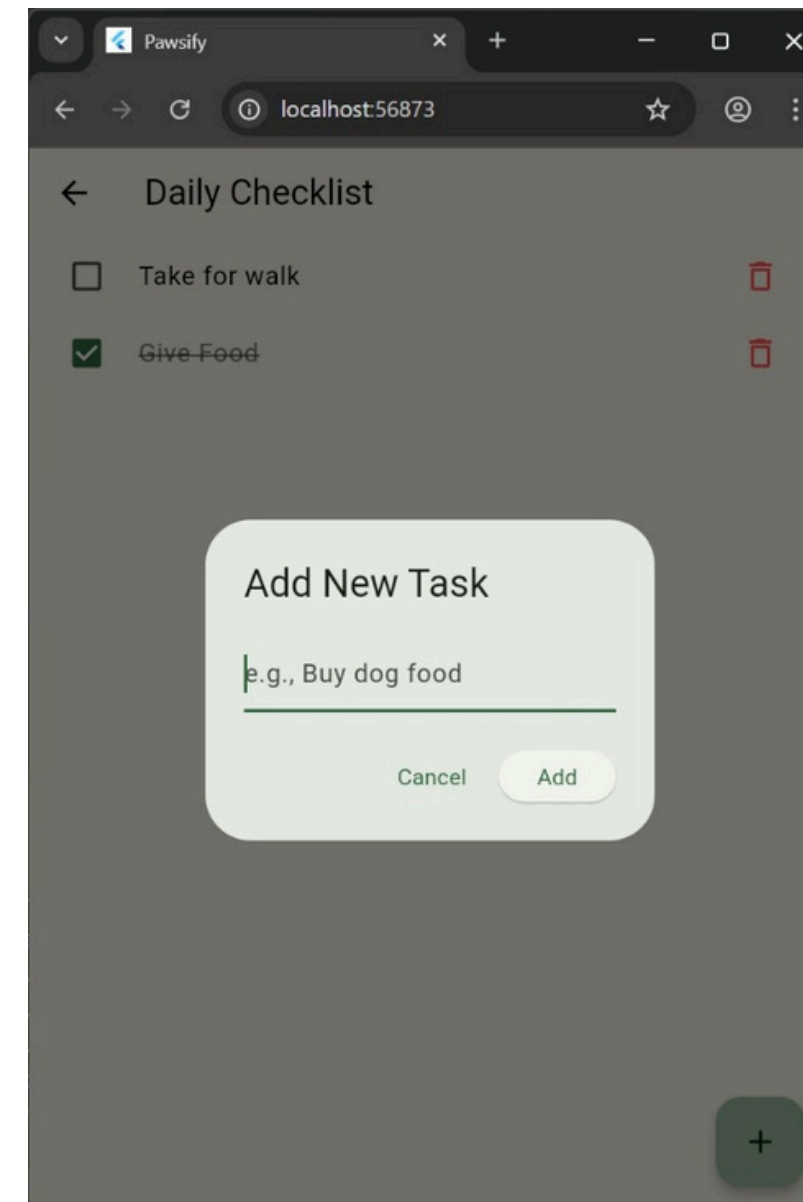
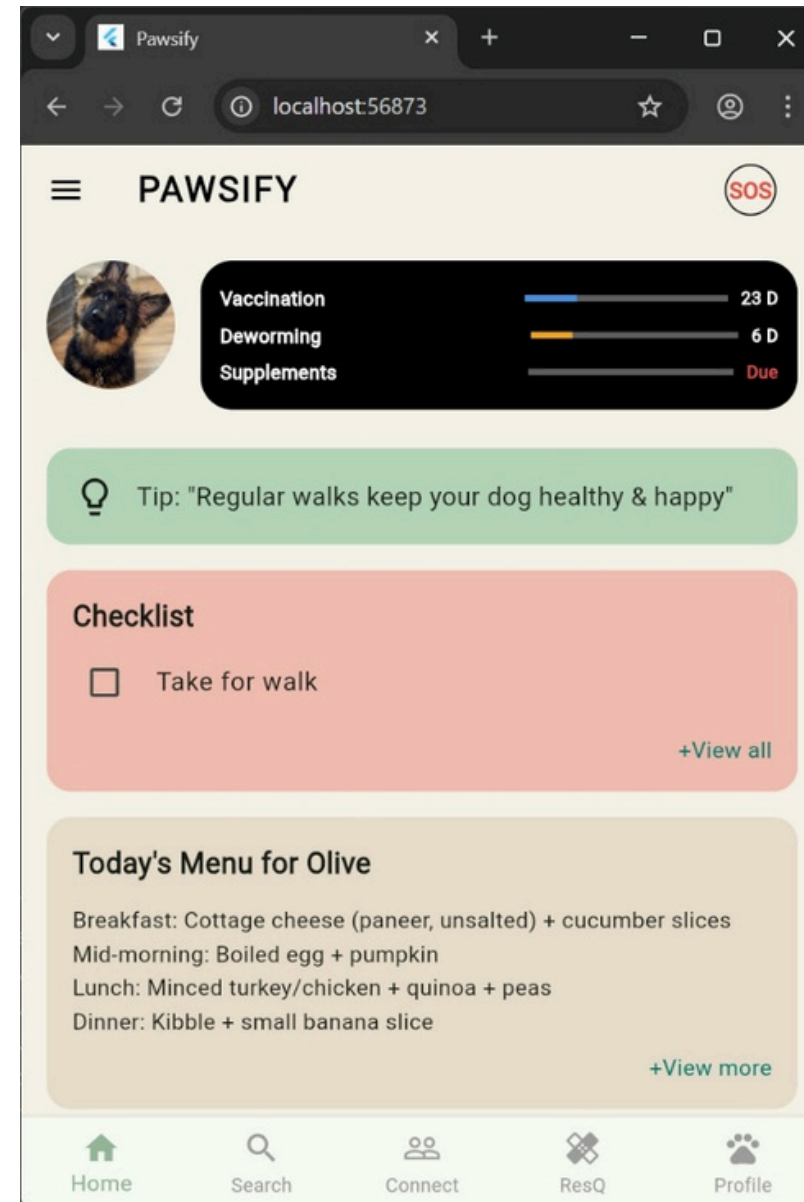
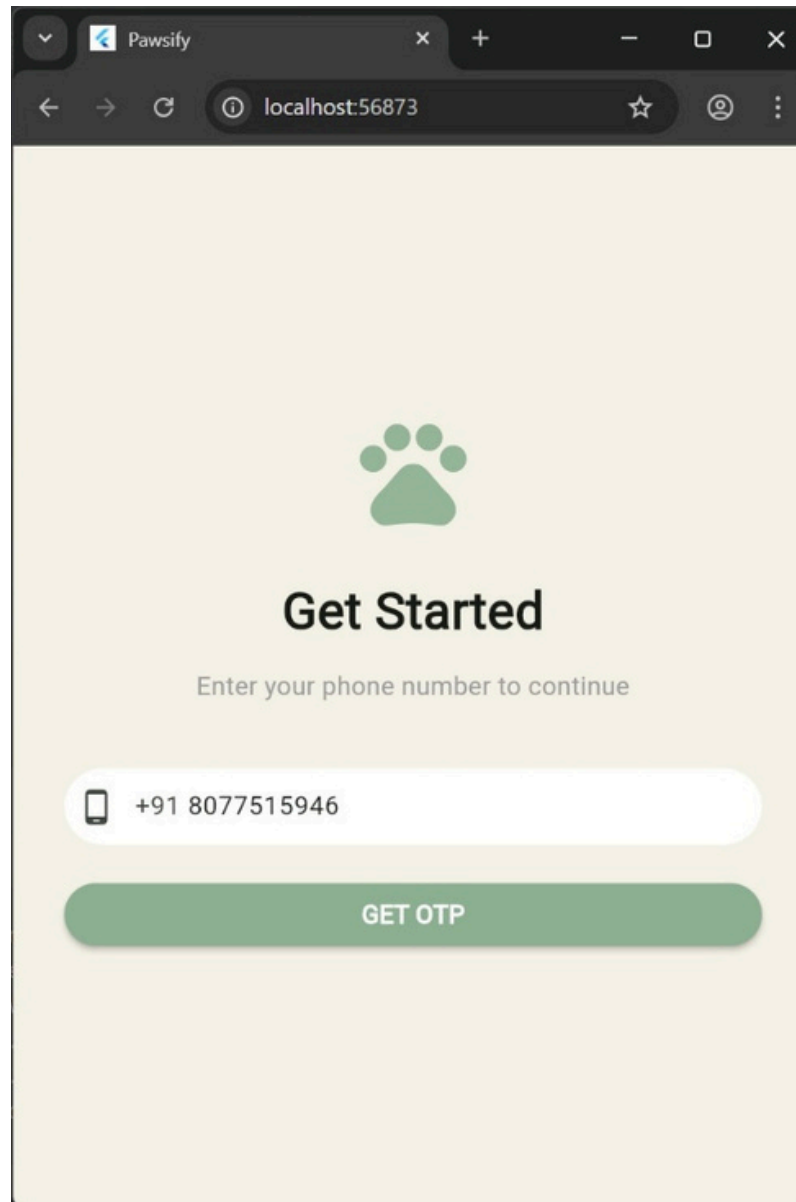
Initial Results

Module / Feature	Test IDs	Status	Remarks
Login & OTP Verification	TC001 – TC002	✓ Pass	Works smoothly; Firebase connected
Pet Profile	TC003	✓ Pass	Add/Edit/View tested successfully
Matchmaking	TC004	✓ Pass	Matches generated accurately
Playdate Finder	TC005	✓ Pass	Able to host an event
Event Hosting	TC006	✓ Pass	Events created and displayed
Search (Map)	TC007	✗ Fail	API integration pending
Donation (ResQ)	TC008	⚠ Partial	Redirects fine; Razorpay not linked
Parent Details	TC009	✓ Pass	Edits saved correctly
Database Integration	TC010	✓ Pass	Firestore data stable
Matchmaking Persistence Bug	TC011	✓ Pass (After Fix)	Session issue resolved

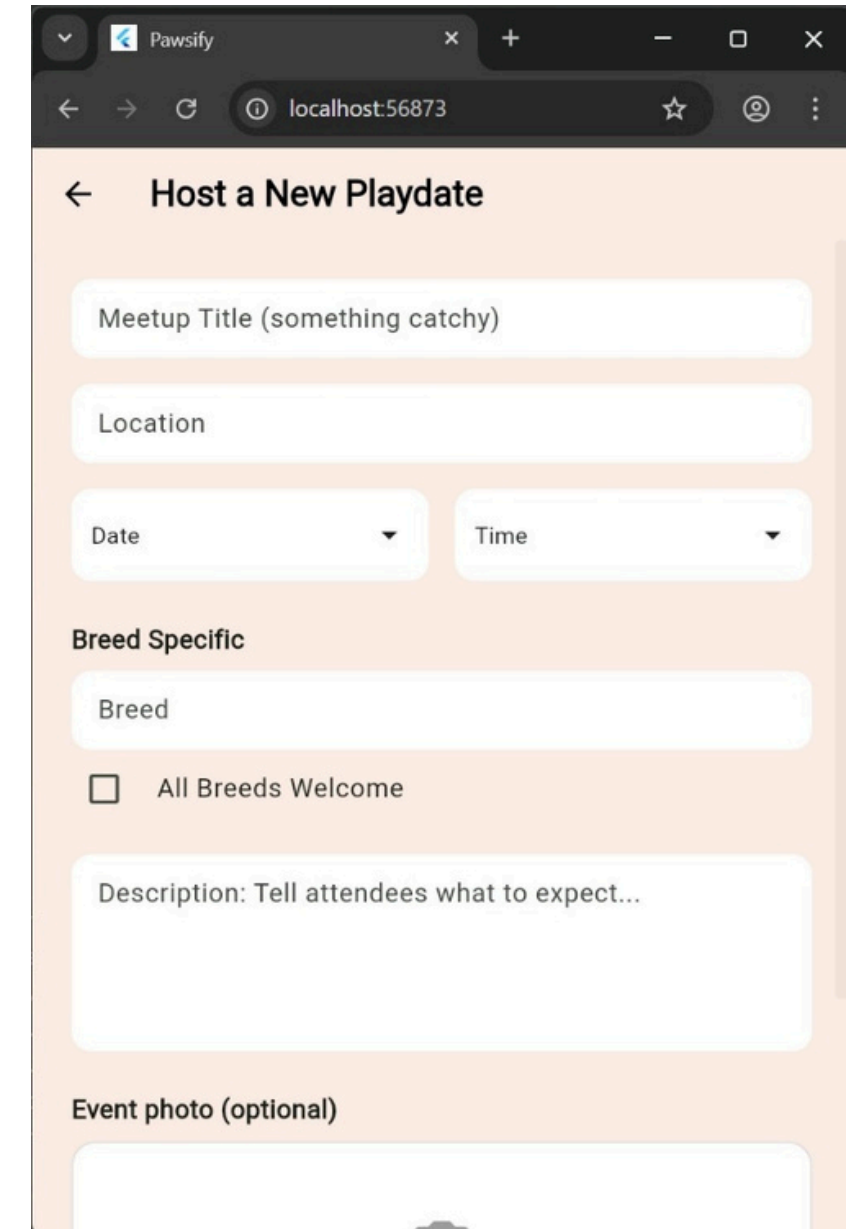
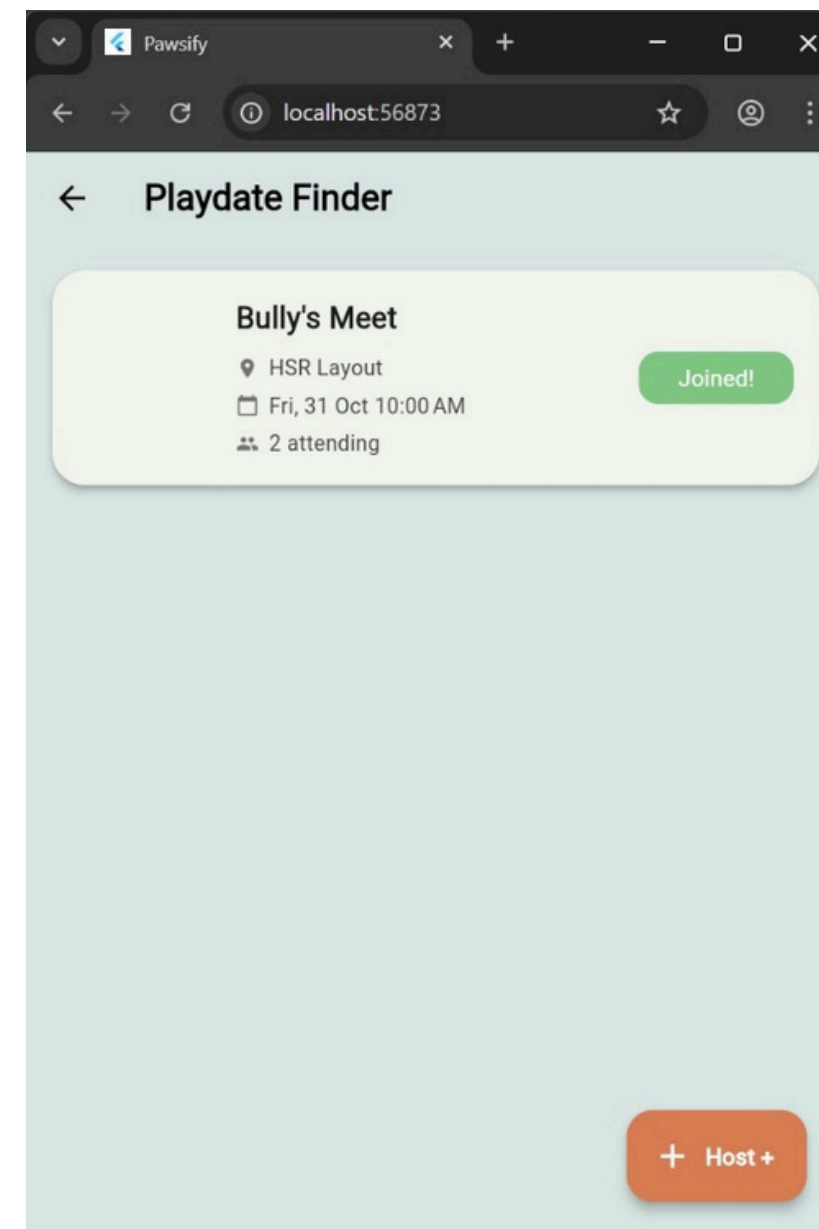
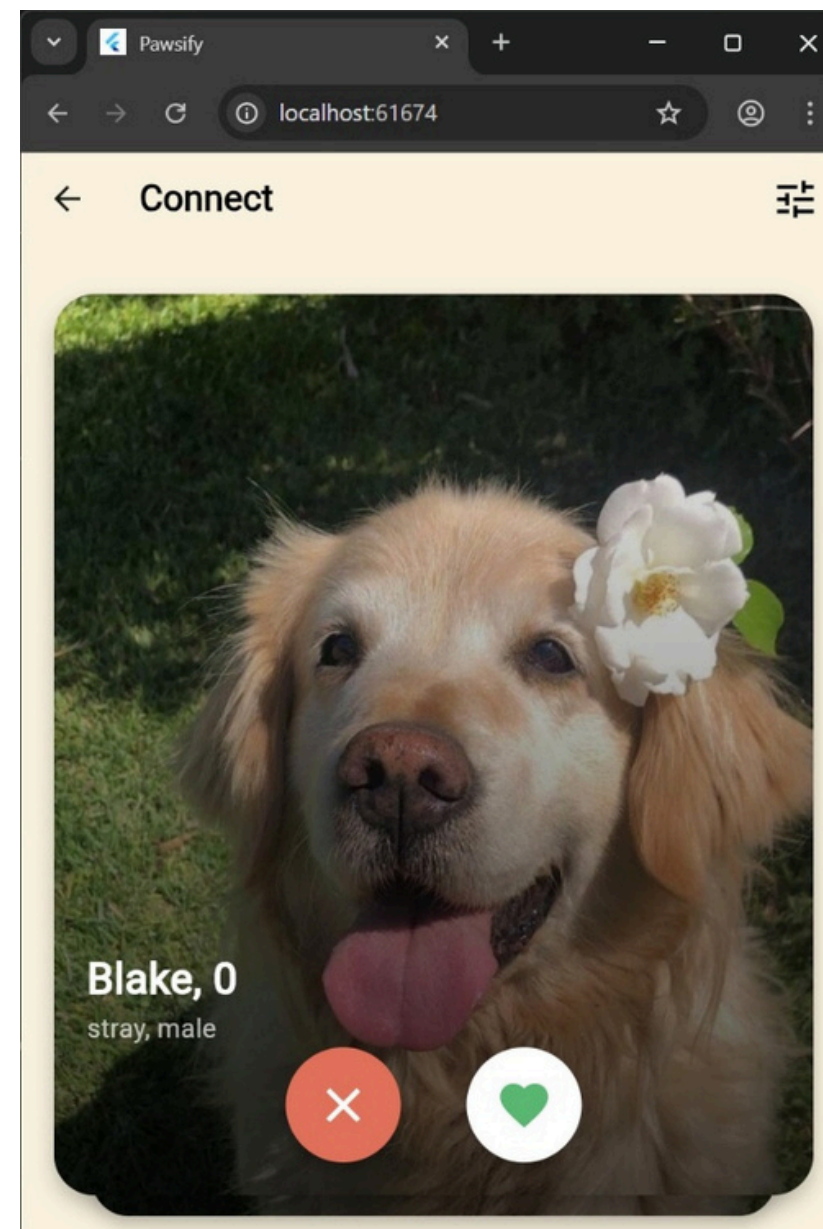
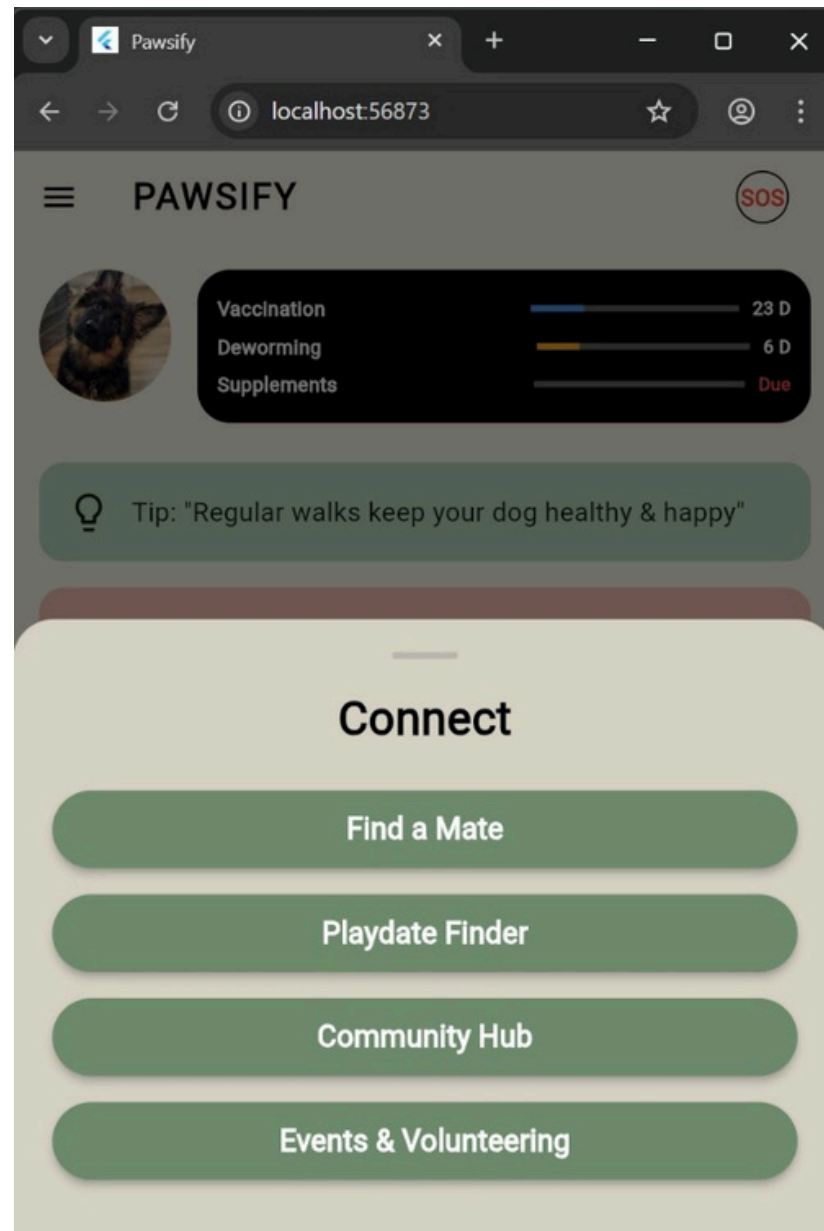
Error Handling

Screen	Frontend (in App)	Backend (Firebase)	How It Handles Error in Code
Login / Phone Number Input (LoginPage.dart)	Uses TextFormField to validate phone number; basic check for empty/length	Firebase throws invalid-phone-number if format is wrong	Simple if (phone.length != 10) validation before calling verifyPhoneNumber(); <u>Snackbar</u> may not be implemented everywhere
Send OTP (OTPPage.dart)	Button disabled while sending; shows progress indicator	Firebase may throw too-many-requests if spammed or throttled	Currently minimal handling — missing explicit try-catch; can be wrapped in on FirebaseAuthException catch (e) with showError("Too many attempts")
OTP Verification (OTPPage.dart)	Checks OTP field not empty before verifying	Firebase may throw invalid-verification-code, session-expired	Partial handling exists; better to wrap in try-catch and show <u>Snackbar</u> or Alert
Enroll Page / Dog Registration (EnrollPage.dart)	Form validation for breed, age, gender	<u>Firestore</u> can fail if user not logged in or network down	Form validation exists; missing error UI on <u>Firestore</u> failure; add try { await set() } catch (e) { showError(...) }
Pet Profile Page (PetProfilePage.dart)	Loads saved pet data; checks for null values	<u>Firestore</u> may return null doc (user not found)	Uses snapshot.hasData check; safe against null; shows "No data found"
Matchmaking Filters (Connect.dart)	Ensures <u>dropdowns</u> (breed, gender) not null	<u>Firestore</u> ignores null query & returns empty result	Uses default values like 'Any'; safe fallback prevents app crash
Matchmaking Swipe (Like/Dislike) (Connect.dart)	Checks that pet card exists before saving like/dislike	<u>Firestore</u> may throw permission-denied if rules fail	try { await saveMatch(...) } catch (e) recommended; not fully implemented
Playdate Host / Finder Pages	Form fields validated (title, date, time)	<u>Firestore</u> rejects writes if not authenticated	Validation exists; <u>Snackbar</u> not consistently shown on write error
Image Upload (Firebase Storage)	Validates if image selected	Firebase throws unauthorised or network-error	Caught implicitly by Firebase; UI fallback uses local asset if URL empty
Global (Any Other Error)	Try-catch in some pages	Firebase logs errors internally	App currently lacks a centralised error handler; recommended to wrap all async Firebase calls with try-catch and show <code>SnackBar(content: Text("Something went wrong"))</code>

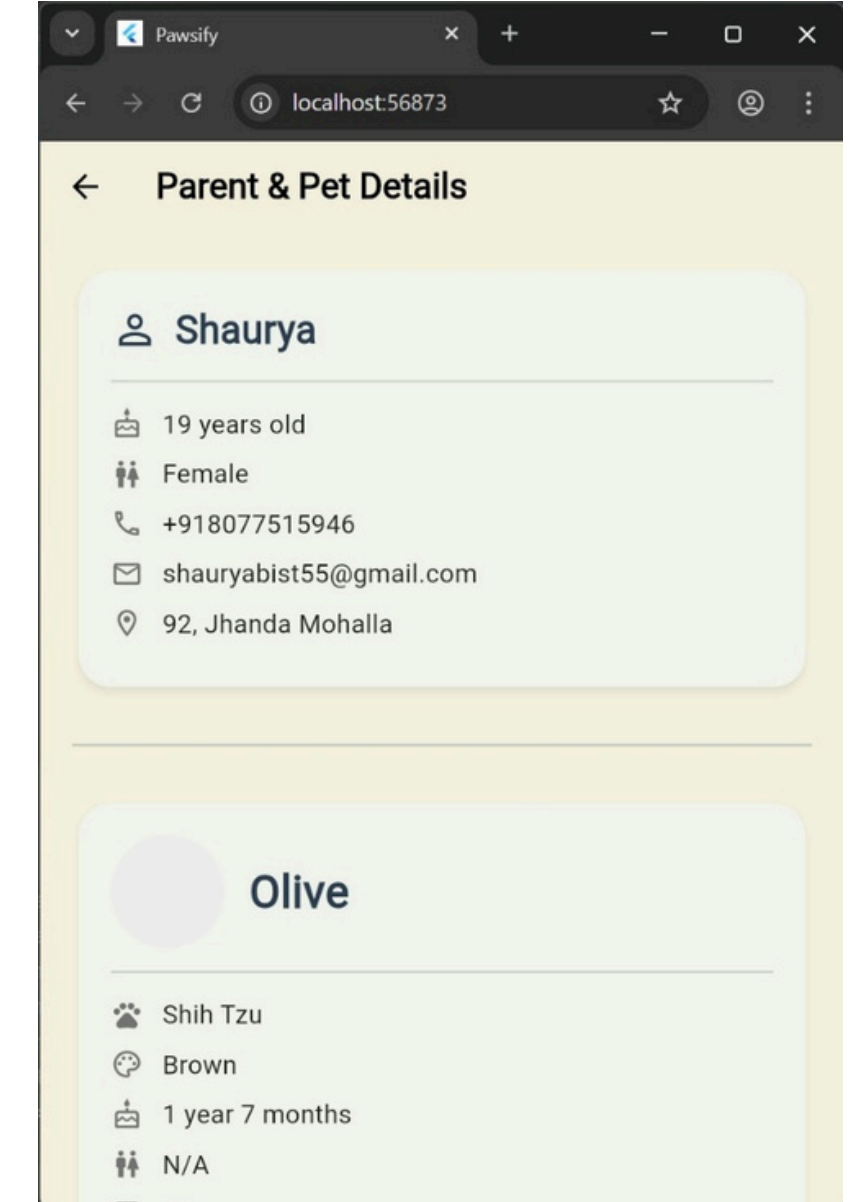
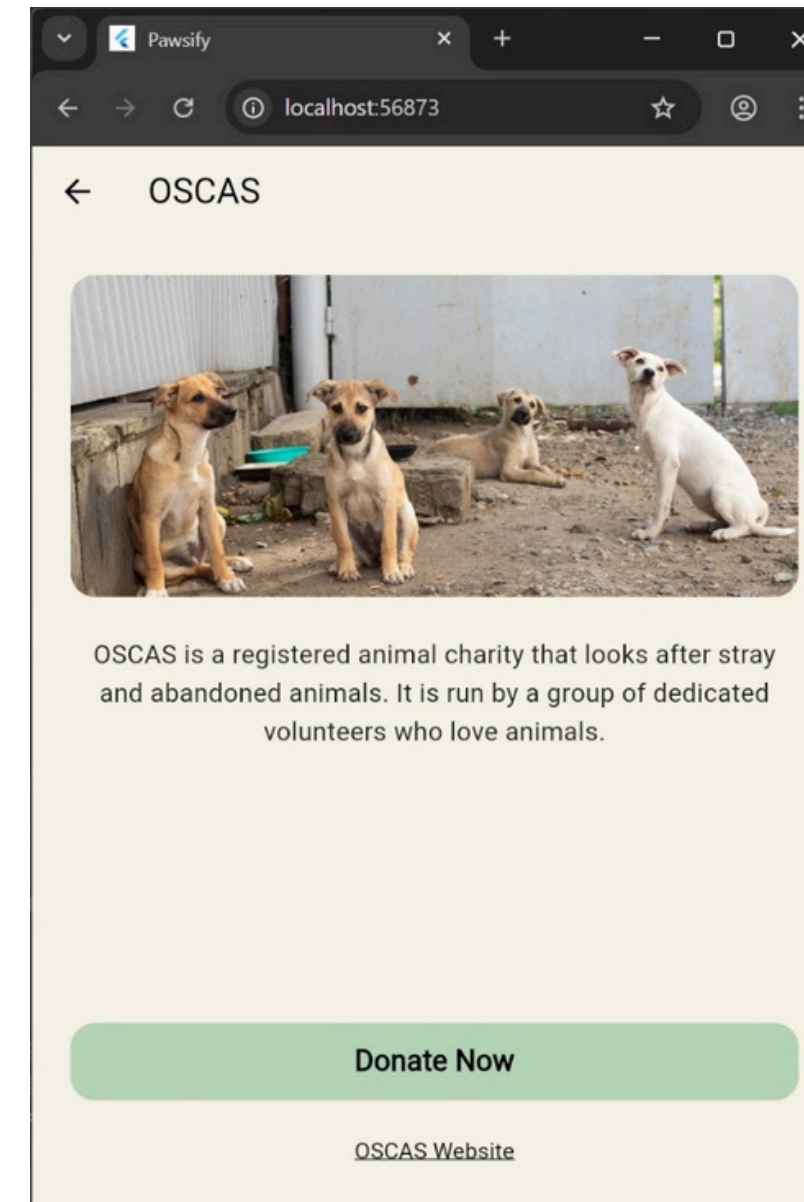
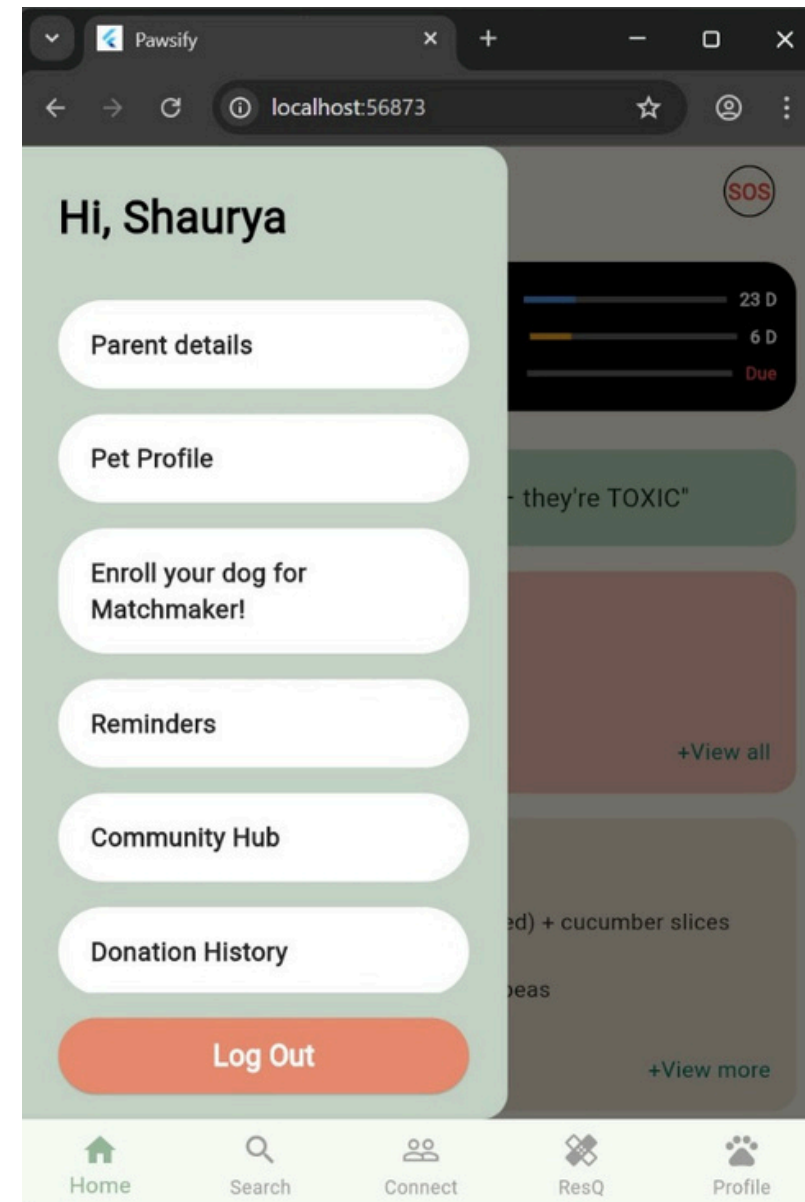
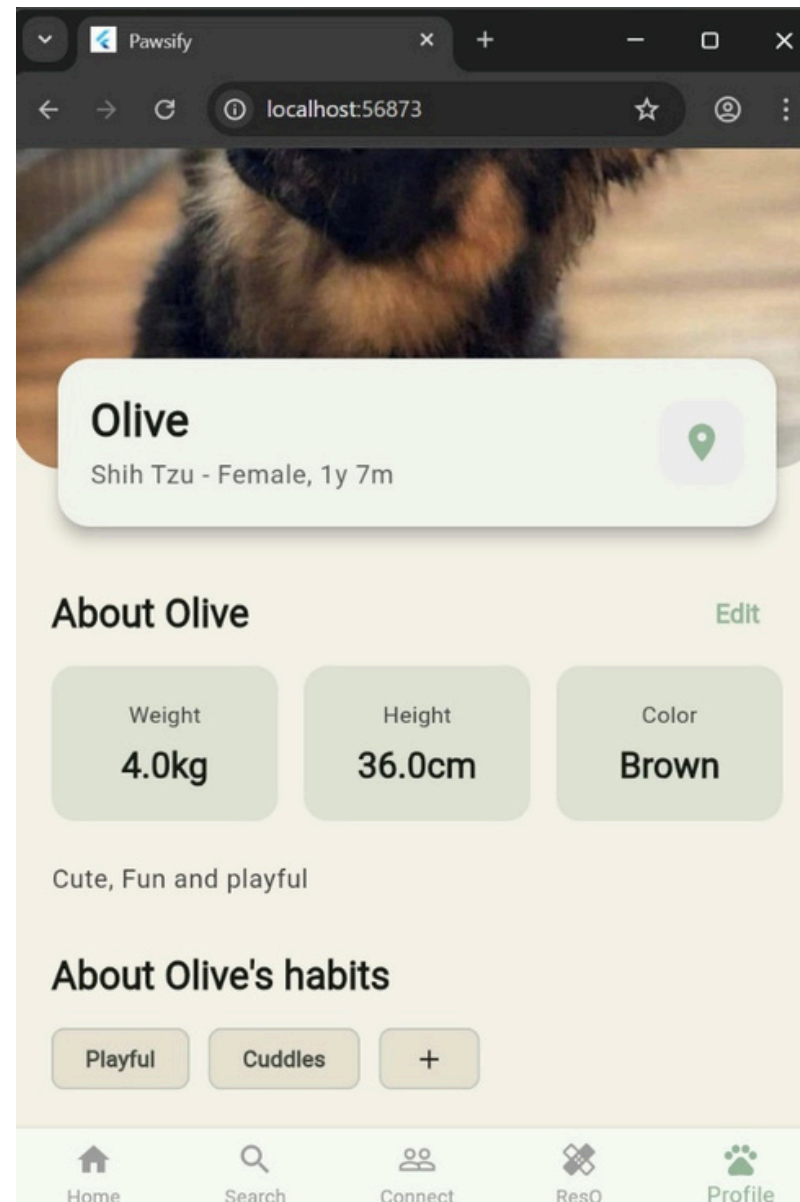
Screenshots / Workflows



Screenshots / Workflows



Screenshots / Workflows



Future Scope

Pillar 1: Core System Expansion & Utility

1. **API Integrations:** Implement external APIs to unlock key functionality:
 - **Payment Gateway** (e.g., Razorpay) for secure transactions and donations.
 - **Mapping Services** for location-based services (vets, events, etc.).
2. **Personalized Pet Care:** Develop data-driven advice:
 - **Breed-Specific Diet & Care Tips.**
Integrate a **Breed Info API** for comprehensive dog data.

Pillar 2: Community & Engagement

- **Dedicated Adoption & Foster Platform:** Build a robust system to connect pets in need with families.
- **Community Hub:** Create a social feed for users to share **posts and stories**, fostering an active pet-owner network.
- **Events & Volunteering:** Feature listings and management for local pet-related activities.
- **Connecting with more organizations:** To help them with adoption and foster manually.

Pillar 3: Platform Refinement & Quality

UI Implementation & Styling: Ensure the final application perfectly **mirrors the approved UI/UX mockups and design assets**, delivering a highly professional and consistent user experience.